

Andrew Manigault, PhD

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PROFESSIONAL SUMMARY

Software engineer with a background in research and expertise in data engineering, UI and statistics. My research background enhances my ability to approach complex problems with a data-driven mindset, making me a strong fit for software engineering roles that require both technical expertise and innovative problem-solving. I possess strong project management skills, having successfully led cross-functional teams and coordinated complex tasks to meet deadlines (e.g., over 25 peer-reviewed publications).

RELEVANT EXPERIENCE

Inductive Health Informatics <i>Software Engineer II</i>	Jan 2023 – Present <i>Atlanta, GA</i>
- Developed containerized tools/applications and software packages aimed at monitoring data quality for the National Syndromic Surveillance Program (NSSP)—a CDC branch that ingests high volume (~88% of emergency departments across the USA) HL7 health data for surveillance tasks. - Improved NSSP operations by 1) easing access/use of data quality tools, 2) enhancing sharing of syndromic surveillance reports, 3) allowing CDC leadership to monitor utilization of tools/resource across the platform. - Successfully maintained and added features to large suite of legacy NSSP web applications. - Refactored legacy application backend to improve load times (~9x on average) and follow object oriented design patterns (e.g., dependency injection) that significantly eased operation and maintenance tasks. - Trained machine learning forecasting models to estimate upcoming changes in disease rates based on seasonality.	
Brown University – Center for the Study of Children at Risk <i>Biostatistician</i>	Sep 2021 – Jan 2023 <i>Providence, RI</i>
- Developed R packages to support research center activities including infant cry acoustic analysis, latent growth curve modeling, multilevel modeling, data cleaning and visualization. - Trained machine learning classifiers to identify neonatal opiate withdrawal syndrome using infant cry acoustics.	
University of California Los Angeles <i>Post-doctoral Research Fellow</i>	Sep 2020 – Dec 2021 <i>Los Angeles, CA</i>
Developed R package to perform multithreaded transcriptomic origin analysis.	

SKILLS

Programming Languages	R Python SQL html CSS JavaScript C# Bash SAS MATLAB
Technologies	Linux Docker/Podman AWS Git Palantir Foundry Unity
Analytic	Bayesian & Frequentist Statistics Machine Learning Multilevel Modeling

NOTABLE PROJECTS

Laboratory Analytics for Situational Awareness & Reporting (LASAR) <i>Full-stack developer</i>	Jan 2025 – Present <i>Atlanta, GA</i>
- Developed a web application to track incoming laboratory data quality issues. - Developed/maintained nightly ETL to refresh SQL tables containing key data quality metrics. - Mentored a junior developer through development of UI, use of version control, and deployment of containerized web applications via Podman/Docker. - Ported application frontend/backend to Palantir Foundry for CDC leadership to evaluate platform capabilities.	
Data Products Analytic Dashboard (DPAD) <i>Full-stack developer</i>	Sept 2024 – Dec 2024 <i>Atlanta, GA</i>
- Developed backend to log user activities across multiple front-end applications via Flask REST API. - Developed containerized web application to summarize user activity across the platform and export activity reports.	
Data Quality On Demand (DQOD) <i>Frontend developer</i>	Sept 2023 – Present <i>Atlanta, GA</i>
Developed web application to generate highly granular/customizable .xlsx data quality reports via stored procedure calls to SQL server DB.	
Publication and Reporting Enterprise for Syndromic Surveillance (PRESS) <i>Frontend developer</i>	Jan 2023 – Present <i>Atlanta, GA</i>
- Developed web application frontend to search/browse, upload, and manage syndromic surveillance reports. - Leveraged Flask REST API backend to allow full automation of app actions (e.g., uploads and updates) via cronjobs.	
Hexpow <i>Mobile Application developer</i>	May 2021 – Jan 2022 <i>Providence, RI</i>
Developed C# mobile game application via Unity and released it on the Google Play store.	

EDUCATION

Doctorate – Psychoneuroendocrinology & Statistics <i>Ohio University</i>	Sep 2016 – May 2020 <i>Athens, OH</i>
Masters of Science - Psychoneuroendocrinology & Statistics <i>Ohio University</i>	Sep 2014 – May 2016 <i>Athens, OH</i>
Bachelor of Science – Psychology <i>Montana State University</i>	Sep 2010 – May 2014 <i>Bozeman, MT</i>